LAPRAK TEKNOLOGI IOT

TUGAS PRAKTIKUM 7

Oleh:

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KELAS : TRI – 3D

Dosen:

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TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

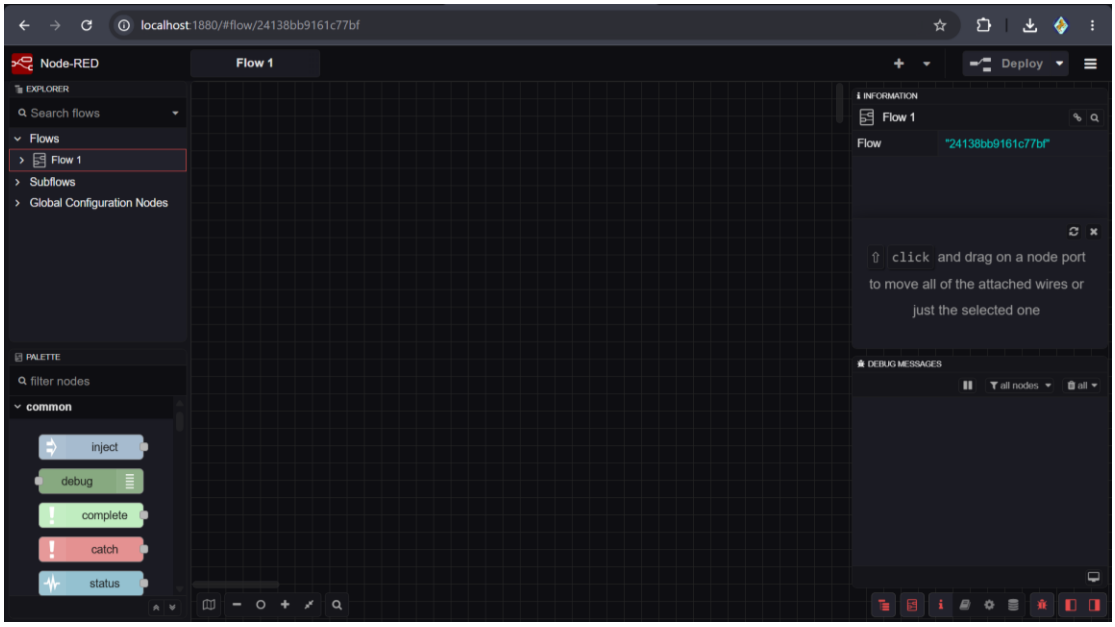
2025/2026

7.5.1 Praktikum 1

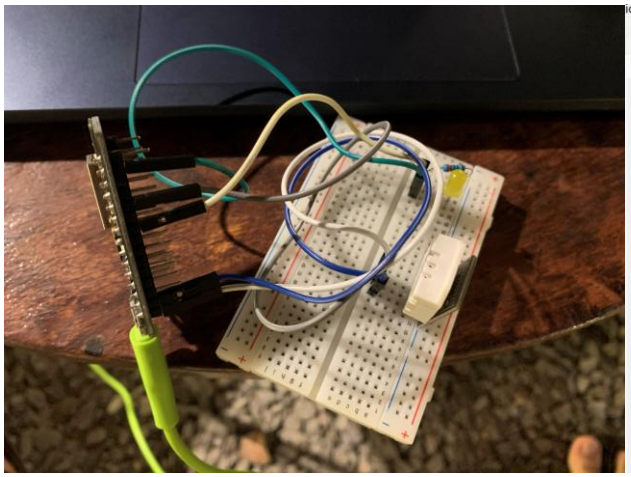
```
Windows PowerShell
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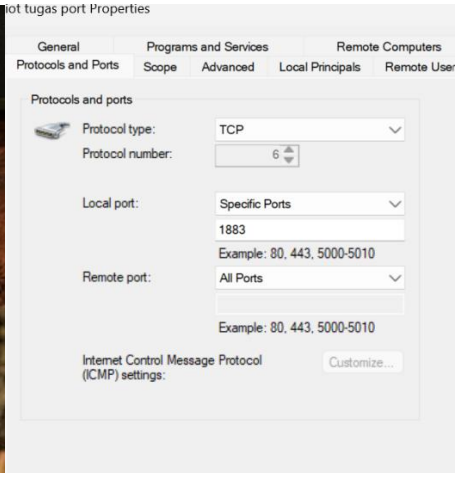
PS C:\Users\ERVAN GUSTI WANJAYA> cd D:\iot-server
PS D:\iot-server> docker compose up -d
[+] up 29/29
 ✔Image nodered/node-red:latest  Pulled                                    34
 ✔Image eclipse-mosquitto:latest Pulled                                     6
 ✔Network iot-server_default     Created                                    0
 ✔Container mqtt-broker          Started                                    1
 ✔Container node-red             Started                                    1
```

```
PS D:\iot-server> docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                    PORTS
                                                                                                                  NAMES
3bfe328fe548   nodered/node-red:latest        "./entrypoint.sh"        54 minutes ago   Up 54 minutes (healthy)   0.0.0.0:1880->1880/tcp,
 [::]:1880->1880/tcp                                                                                              node-red
07b1679ad0f4   eclipse-mosquitto:latest       "/docker-entrypoint.…"   54 minutes ago   Up 15 seconds             0.0.0.0:1883->1883/tcp,
 [::]:1883->1883/tcp, 0.0.0.0:9001->9001/tcp, [::]:9001->9001/tcp                                                 mqtt-broker
PS D:\iot-server>
```



7.5.2 Praktikum 2

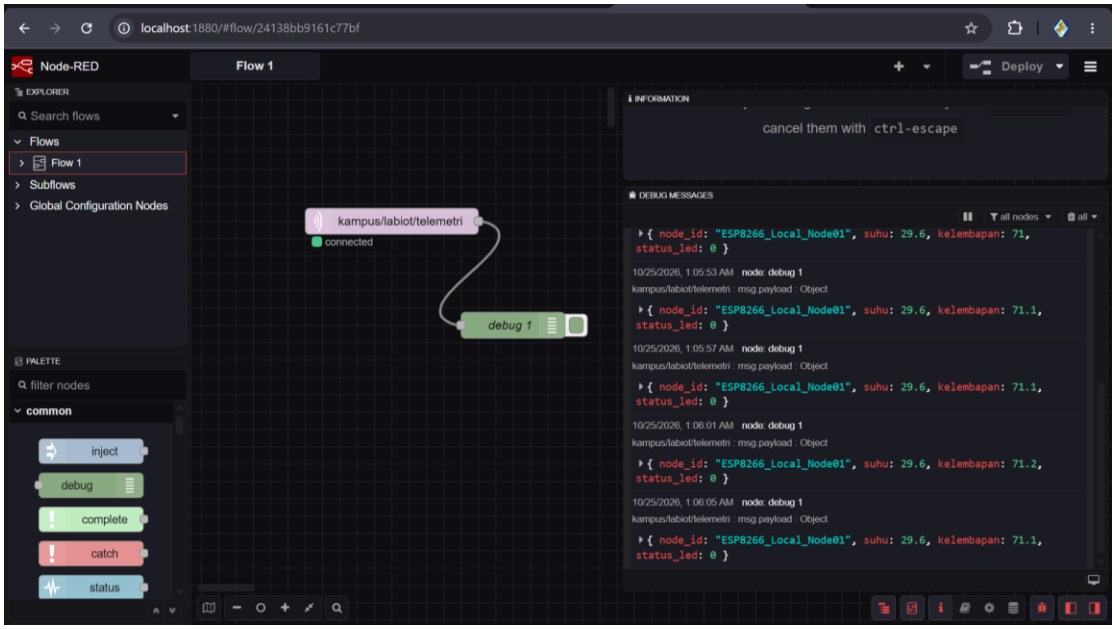


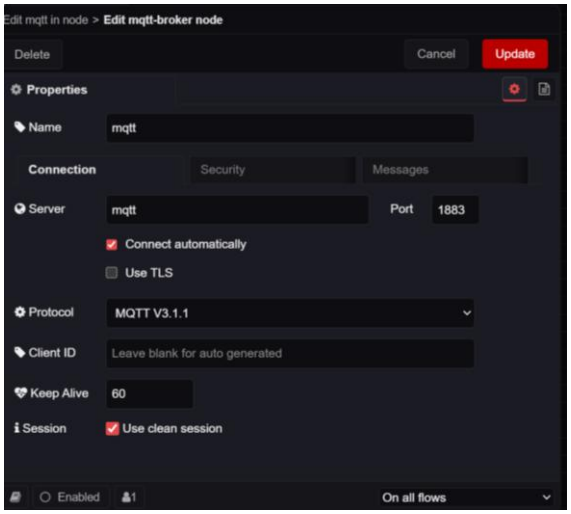


```
[MQTT] Menghubungkan ke Broker Lokal (10.85.111.148)... Gagal (rc=-2). Coba lagi dalam
[MQTT] Menghubungkan ke Broker Lokal (10.85.111.148)... Gagal (rc=-2). Coba lagi dalam
[MQTT] Menghubungkan ke Broker Lokal (10.85.111.148)... Gagal (rc=-2). Coba lagi dalam
[MQTT] Menghubungkan ke Broker Lokal (10.85.111.148).....
[Wi-Fi] Berhasil Terhubung!
[Wi-Fi] IP NodeMCU: 10.85.111.173
[MQTT] Menghubungkan ke Broker Lokal (10.85.111.148)... Terhubung!
```

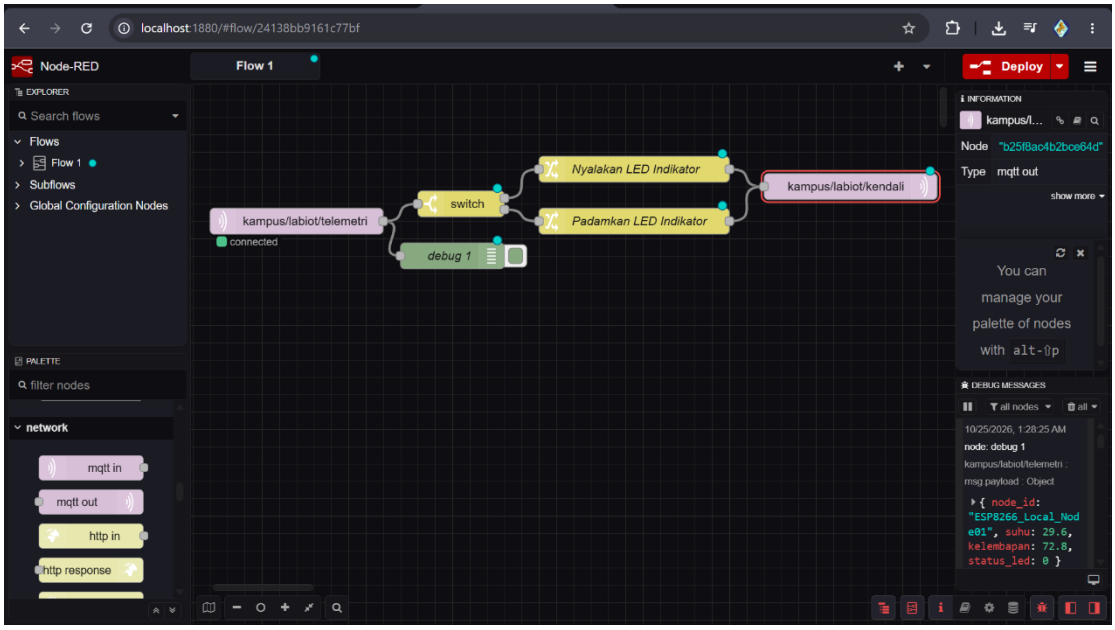
```
[MQTT] Berlangganan ke topik: kampus/labiot/kendali
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":77.60,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":77.70,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":77.80,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":77.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":78.20,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":27.60,"kelembapan":78.40,"status_led":0}
```

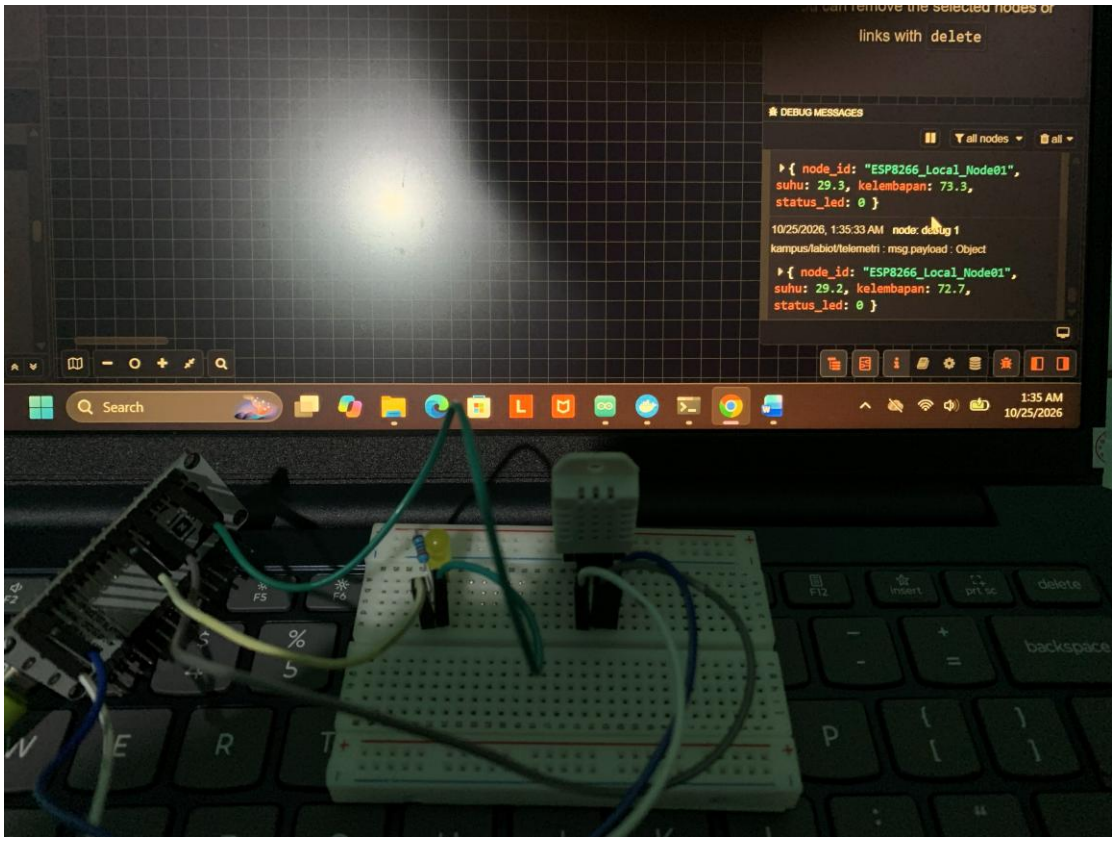
7.5.3 Praktikum 3:

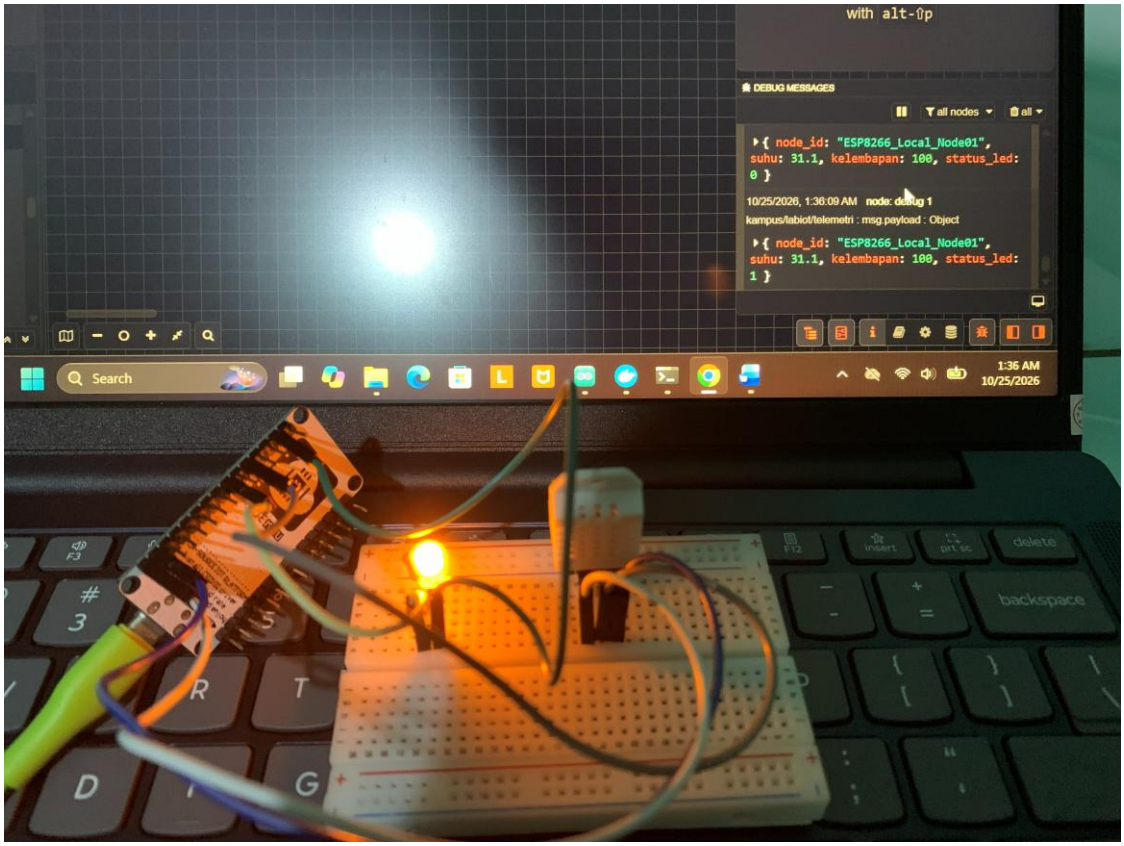




7.5.4 Praktikum 4








```
----------------------------------------
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD]         : {"led":1}
--> [AKTUATOR] Status LED diubah: MENYALA (ON)
----------------------------------------
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":31.00,"kelembapan":99.90,"status_led":1}

----------------------------------------
[INSTRUKSI MASUK] Topik: kampus/labiot/kendali
[PAYLOAD]         : {"led":0}
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
----------------------------------------
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":30.90,"kelembapan":89.30,"status_led":0}
```

7.6 Latihan

Latihan 1: Jaringan Internal Docker & Service Discovery

- Penggunaan localhost di dalam Node-RED merujuk pada ruang lingkup kontainer itu sendiri, bukan pada laptop *host* atau kontainer Mosquitto, sehingga permintaan koneksi ke port tersebut ditolak.
- Docker Compose menciptakan *bridge network* khusus yang menghubungkan para kontainer. DNS internal Docker secara otomatis menterjemahkan nama layanan seperti mqtt menjadi alamat IP internal kontainer, mempermudah pelacakan alamat dinamis.

Latihan 2: Penanganan Konektivitas Jaringan

- IP ESP8266 (192.168.43.102) dan laptop (192.168.1.15) berada di segmen jaringan (*subnet*) yang berbeda, memblokir lalu lintas lokal dan permintaan *broadcast* MQTT secara langsung.
- Hubungkan kedua perangkat ke *Access Point* yang sama, salah satunya dengan mengaktifkan fitur Mobile Hotspot pada Windows dan menyambungkan NodeMCU ke SSID hotspot laptop tersebut.

Latihan 3: Implementasi Filter Histeresis

```
let ledState = context.get("ledState") || 0;
let suhu = msg.payload.suhu;

if (suhu > 31.5) {
    ledState = 1;
} else if (suhu < 30.0) {
    ledState = 0;
}

context.set("ledState", ledState);
msg.payload = { "led": ledState };
return msg;
```

Latihan 4: Volume Mount & Persistensi Data

- Seluruh rancangan kanvas visual *flow* Node-RED dan basis data akan musnah secara permanen karena *file system* bawaan kontainer bersifat sementara (*ephemeral*) saat siklus hidupnya berakhir.
- *Bind Mount* menambatkan jalur direktori laptop pengguna secara spesifik ke dalam kontainer, sedangkan *Named Volume* dikelola sepenuhnya oleh mesin Docker di ruang penyimpanan yang terisolasi dari *host*.

7.7 Tugas

1. Membuat Berkas Kredensial Terenkripsi di Dalam Kontainer:

```
PS D:\iot-server> docker exec -it mqtt-broker mosquitto_passwd -c /mosquitto/config/password.txt admin
Password:
Reenter password:
Adding password for user admin
```



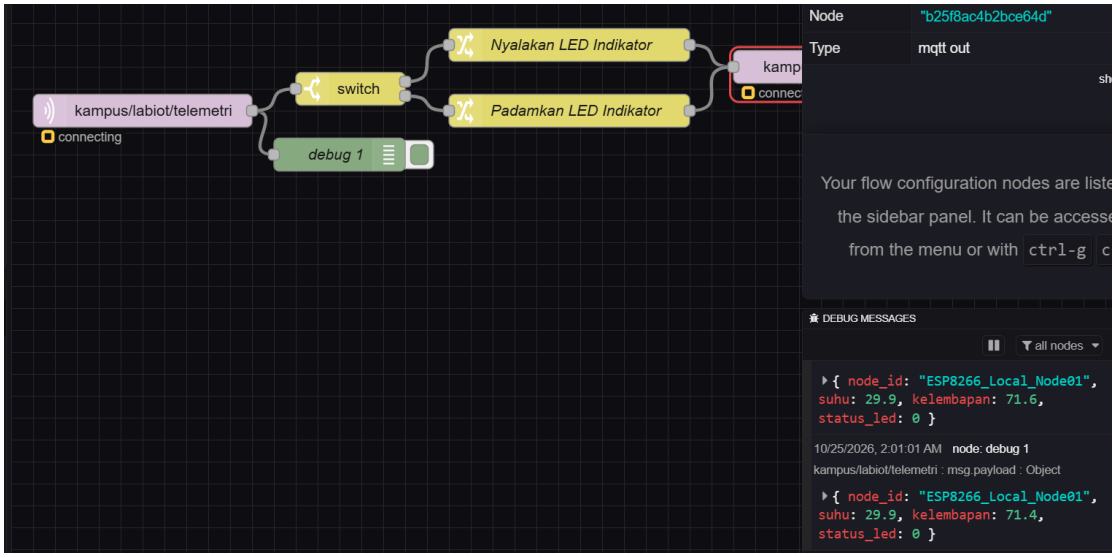
2. Memperbarui Konfigurasi Kebijakan Mosquitto:

```
# Perizinan Akses Awal (Tanpa Password)
allow_anonymous false
password_file /mosquitto/config/password.txt
```

3. Memuat Ulang Layanan Broker:

```
PS D:\iot-server> docker restart mqtt-broker
mqtt-broker
```

4. Analisis Dampak Kegagalan Sistem (Observasi Masalah):

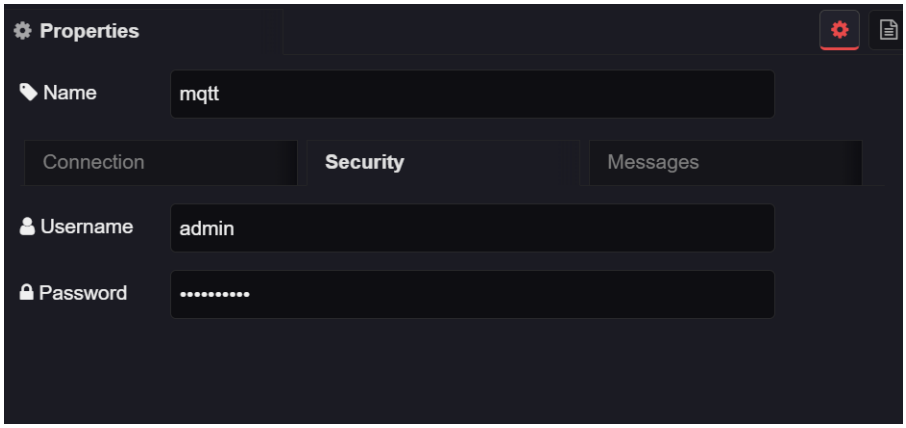


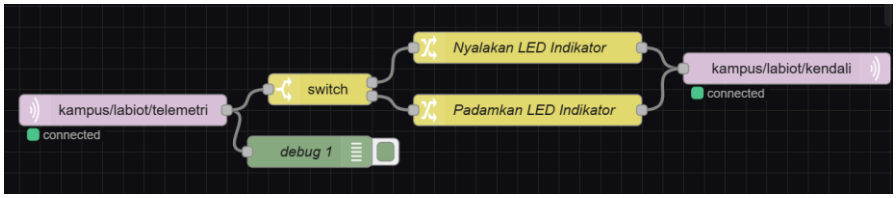

```
--> [AKTUATOR] Status LED diubah: PADAM (OFF)
--------------------------------------
[MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-4). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-2). Coba lagi dalam 4 detik...
[MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-2). Coba lagi dalam 4 detik...
```

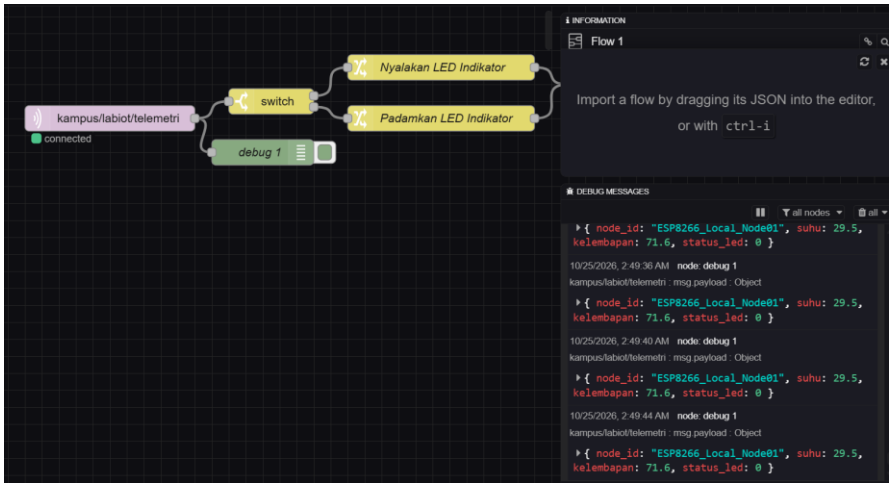
5. Rekayasa Perbaikan Sistem (Penyelesaian Tugas):

```
void reconnectMQTT() {
  while (!client.connected()) {
    Serial.print("[MQTT] Menghubungkan ke Broker Lokal (");
    Serial.print(mqttServer);
    Serial.print(")...");

    // Menambahkan username "admin" dan password "rahasia123"
    if (client.connect(clientId, "admin", "rahasia123")) {
      Serial.println(" Terhubung!");
      // Segera daftarkan subskripsi topik kendali
      client.subscribe(topicSubCmd);
      Serial.printf("[MQTT] Berlangganan ke topik: %s\n", topicSubCmd);
    } else {
      Serial.printf(" Gagal (rc=%d). Coba lagi dalam 4 detik...\n", client.state());
      delay(4000);
    }
  }
}
```







```
MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Gagal (rc=-4). Coba lagi dalam 4 detik...
MQTT] Menghubungkan ke Broker Lokal (192.168.1.15)... Terhubung!
MQTT] Berlangganan ke topik: kampus/labiot/kendali
TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":29.50,"kelembapan":71.70,"status_led":0}

---------------------------------------
INSTRUKSI MASUK] Topik: kampus/labiot/kendali
PAYLOAD]         : {"led":0}
-> [AKTUATOR] Status LED diubah: PADAM (OFF)
---------------------------------------
TELEMETRI TERKIRIM] {"node_id":"ESP8266_Local_Node01","suhu":29.50,"kelembapan":71.70,"status_led":0}
```